

Value Extinction Conditions Module (VXCM)

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Canonical Definition

Value Extinction Conditions Module (VXCM) is the first Core Module of the Value Extinction Standard (VXSS) within VFM® — Value Flow Mechanisms Architecture.

VXCM governs the identification, definition, qualification, and contextualization of conditions indicating that previously existing value may be approaching or entering a terminal loss of recoverability within its governed lifecycle.

The Module establishes whether conditions exist that justify moving a value from ordinary degradation, dormancy, impairment, inaccessibility, nullification, or zero-value governance into formal Value Extinction consideration.

VXCM does not determine that Value Extinction has occurred.

Its task is narrower:

to establish whether sufficiently material extinction conditions exist to justify evaluating whether the relevant value is approaching the boundary beyond which recovery of that same value may no longer remain legitimate.

Architectural Position

VXCM is the first operational control point within VXSS.

It receives a value whose lifecycle may already include:

Value Change

→ Value Degradation

→ Value Nullification

→ ZVA™ — Zero Value Asset™

→ failed, unavailable, or exhausted restoration pathways

or another condition capable of raising a legitimate extinction question.

VXCM does not begin by assuming extinction.

It begins by asking whether the observed circumstances constitute:

temporary or recoverable conditions

or

credible extinction conditions requiring terminal-boundary evaluation.

The output of VXCM provides the governed basis for the next stage of extinction governance.

Internal Flow

Value Under Extinction Consideration → Current Value Condition Identification → Potential Extinction Condition Identification → Condition Classification → Condition Materiality Assessment → Recoverability Relevance Assessment → Condition Persistence Assessment → Condition Dependency Assessment → Extinction Condition Qualification → Governed Extinction Conditions → Value Extinction Threshold Assessment

Operational Space

VXCM governs the space between:

a value experiencing serious impairment, loss, zero state, or restoration difficulty

and

a value for which legitimate extinction conditions have been established.

Its operational space includes:

potential extinction-condition identification; condition definition; condition classification; condition materiality; condition persistence; condition reversibility;

recoverability relevance; restoration-pathway relevance; terminal-condition indicators; condition dependencies; compound extinction conditions; contextual extinction conditions; dimensional extinction conditions; temporal extinction conditions; uncertainty surrounding extinction conditions; qualification of extinction conditions.

Operational Scope

VXCM may govern extinction conditions affecting value associated with:

physical assets; industrial assets; infrastructure; digital assets; data; information; intellectual property; technologies; products; services; productive capabilities; operational capabilities; organizational capabilities; economic relationships; contractual or economic rights; systems; platforms; knowledge structures; other governable Value Objects.

The Module remains value-focused.

It does not independently determine whether a physical, legal, technological, economic, operational, or informational event occurred.

Those facts may originate from other governed systems.

VXCM determines:

whether those conditions are materially relevant to the continued recoverability of the governed value.

Extinction Condition Principle

An Extinction Condition is a governed condition capable of materially contributing to the loss of recoverability of previously existing value.

An Extinction Condition does not itself prove extinction.

It establishes:

a condition that must be considered when determining whether the relevant value remains recoverable.

This distinction is fundamental:

Extinction Condition $\neq$ Value Extinction.

Condition $\neq$ Determination

A physical asset may be destroyed.

A legal right may expire.

A dataset may lose critical information.

A technology may become incompatible with its operating environment.

A market relationship may disappear.

Each may constitute a potentially material extinction condition.

But VXCM does not automatically convert the condition into:

Value Extinct.

The relevant value may persist through another dimension, context, capability, relationship, or restoration pathway.

Therefore:

Terminal-Looking Event $\neq$ Automatic Value Extinction.

Extinction Conditions Must Be Value-Specific

VXCM requires every extinction condition to remain connected to the specific governed value being evaluated.

For example:

A machine becoming technologically obsolete may constitute a serious condition for:

productive manufacturing value

while having little negative effect on:

historical value

or

component value.

Similarly, deletion of a software platform may eliminate one operational value while preserving intellectual property, data, knowledge, or transferable technical value.

Therefore:

Extinction Conditions Are Value-Specific, Not Merely Object-Specific.

Condition Classification

Extinction conditions may arise through different mechanisms.

VXCM may classify conditions as:

Physical Conditions

Conditions affecting the physical basis required for the value to exist or be recovered.

Examples may include:

irreversible destruction;

structural failure; material exhaustion; permanent physical incompatibility; unrecoverable component loss.

Functional Conditions

Conditions affecting the capability through which the relevant value was produced.

Examples may include:

irreversible loss of functionality; inability to perform the relevant function; permanent loss of required capability; functional incompatibility.

Informational Conditions

Conditions affecting information required for the value to remain meaningful or recoverable.

Examples may include:

irreversible data loss; missing metadata; destroyed decoding information; unrecoverable provenance; loss of required contextual information.

Technological Conditions

Conditions affecting the technological basis required for recovery or continuation.

Examples may include:

permanent incompatibility; disappearance of essential technological dependencies; irreversible architecture discontinuity; loss of required reconstruction capability.

Legal or Rights-Dependent Conditions

Conditions affecting value that depends materially upon a legal or rights-based relationship.

Examples may include:

irreversible loss of a necessary right; permanent termination of a value-dependent entitlement; expiry where the value cannot continue independently.

VXCM does not determine the legal status itself.

It governs the resulting relevance to value recoverability.

Operational Conditions

Conditions affecting the operational capability required for value recovery.

Examples may include:

permanent loss of essential operating capability; irreversible infrastructure discontinuity; permanent inability to recreate required operating conditions.

Relational Conditions

Conditions affecting value that exists through a specific relationship.

Examples may include:

irreversible loss of the relationship required for the value; permanent separation of dependent elements; disappearance of the context in which the value relationship can exist.

Temporal Conditions

Conditions in which time materially affects recoverability.

Examples may include:

irreversible expiry; loss of a time-dependent opportunity; permanent disappearance of a temporal window necessary for the relevant value.

Contextual Conditions

Conditions in which the original value context itself has ceased to support the relevant value relationship.

Contextual extinction must remain bounded to that context.

Contextual Extinction Condition $\neq$ Universal Extinction Condition.

Single Condition $\neq$ Sufficient Extinction Basis

Value Extinction may depend upon one decisive condition.

But in many cases, extinction emerges through several interacting conditions.

For example:

obsolete technology

alone may not eliminate recoverability.

But:

obsolete technology

irrecoverable technical documentation

unavailable components

loss of necessary operational infrastructure

may together create a materially different extinction basis.

VXCM therefore supports:

Compound Extinction Conditions.

Compound Extinction Conditions

A Compound Extinction Condition exists where multiple individually non-terminal conditions collectively create a credible threat to recoverability.

VXCM must therefore evaluate:

condition interaction; condition dependence; cumulative effect; whether one condition amplifies another; whether removal of one condition would restore recoverability; whether the combination creates terminal characteristics absent from the individual conditions.

This prevents extinction governance from evaluating complex value loss through isolated indicators only.

Condition Materiality

Not every negative condition is materially relevant to extinction.

VXCM requires the significance of a condition to be evaluated against the governed value.

A condition is materially relevant where it meaningfully affects:

the basis of the value; the ability of the value to persist; restoration potential; recoverability; continuity of the value relationship; the possibility of returning the value through ReMonet™.

Minor impairment should not be elevated into extinction governance merely because it is observable.

Condition Persistence

VXCM distinguishes between:

Temporary Condition

and

Persistent Condition.

A condition may temporarily prevent value recovery without meaningfully threatening long-term recoverability.

Examples include:

temporary infrastructure outage; temporary legal restriction; temporary market absence; temporary technological limitation; temporary lack of capital; temporary access failure.

Persistence strengthens extinction relevance only where continued duration materially affects the possibility of recovering the same value.

Persistence ≠ Permanence

A long-lasting condition is not automatically permanent.

Therefore:

Persistent Condition ≠ Extinction Condition by Default.

A value may remain dormant or inaccessible for decades while preserving meaningful restoration potential.

VXCM requires the condition's relationship to recoverability to be established rather than inferred from duration alone.

Reversibility Principle

A condition that is readily reversible generally provides a weaker basis for extinction consideration than one that cannot reasonably be reversed.

VXCM may distinguish:

Reversible

Conditionally Reversible

Unknown Reversibility

Practically Irreversible

Structurally Irreversible

conditions.

However:

Irreversible Condition $\neq$ Automatic Value Extinction.

The condition may affect only one component or one value dimension.

The relevant value may remain recoverable through another pathway.

Recoverability Relevance

Every material extinction condition should be evaluated against one central question:

How does this condition affect the ability of the same governed value to return?

The condition may:

have no material effect on recoverability; reduce recoverability; materially constrain recoverability; remove one restoration pathway; remove multiple restoration pathways; leave only conditional restoration; appear to eliminate all legitimate restoration pathways.

VXCM identifies this relationship.

It does not make the final extinction determination.

Restoration Pathway Loss

Loss of a restoration pathway may constitute a material extinction condition.

But:

Loss of One Restoration Pathway $\neq$ Loss of Recoverability.

Multiple pathways may exist.

For example, an industrial asset may no longer be repairable through original components but may remain recoverable through:

reconstruction; remanufacturing; technological conversion; substitution; digital reconstruction; alternative production methods.

VXCM must therefore avoid confusing:

one unavailable recovery method

with

absence of legitimate recoverability.

Failed ReMonet™ as a Condition

A failed ReMonet™ process may become relevant evidence within extinction-condition governance.

But the correct relationship is:

Failed ReMonet™ → Potential Extinction-Relevant Condition

not:

Failed ReMonet™ → Value Extinction.

The failure may reveal:

deeper structural impairment; missing restoration capability; underestimated dependencies; previously unknown information loss; technological barriers.

It may therefore strengthen the extinction basis without independently determining extinction.

Zero Value as a Condition

A zero-value state may increase the relevance of extinction governance.

But:

ZVA™ ≠ Extinction Condition by Definition.

A ZVA™ may possess substantial restoration potential.

Zero value therefore indicates:

current absence of positive governed value

not necessarily:

terminal loss of recoverability.

VXCM must preserve this distinction before the extinction threshold is considered.

Dormancy as a Condition

Dormancy alone does not constitute Value Extinction.

Dormancy becomes extinction-relevant only where associated conditions materially threaten the possibility of recovery.

Therefore:

Dormancy ≠ Extinction Condition by Default.

Obsolescence as a Condition

Obsolescence may be extinction-relevant.

It is not automatically terminal.

Obsolete assets, technologies, data, knowledge, or capabilities may:

be repurposed; be modernized; regain market relevance; gain historical value; support new applications; become strategically useful.

VXCM therefore treats obsolescence as:

a condition requiring value-specific interpretation

rather than:

proof of extinction.

Economic Infeasibility

A value may currently cost more to restore than its expected economic return.

This may be materially relevant to organizational decisions.

But economic infeasibility must remain distinguishable from fundamental non-recoverability.

Therefore:

Economically Irrational to Restore $\neq$ Necessarily Extinct.

Where the governed value itself is economic, however, economic conditions may form part of the relevant extinction basis.

The relationship must be explicitly defined.

Condition Dependency

Extinction conditions may depend upon one another.

For example:

loss of software compatibility

may become material only because:

the original source code is unavailable

and

the hardware platform can no longer be reconstructed.

VXCM requires materially dependent conditions to remain connected rather than recorded as unrelated observations.

Condition Uncertainty

Extinction conditions may themselves be uncertain.

VXCM should preserve uncertainty concerning:

whether the condition actually exists; whether it is permanent; whether it is reversible; whether it affects the relevant value; whether alternative restoration pathways remain; whether future capability could materially change the condition.

Where uncertainty remains material:

Uncertain Condition $\neq$ Qualified Terminal Condition.

Extinction Condition Qualification

A potential condition becomes a Governed Extinction Condition when its relationship to the relevant value and recoverability is sufficiently established for downstream extinction-threshold assessment.

Qualification should establish, where materially relevant:

the condition; affected value; affected Value Object; affected value dimension; relevant context; condition origin; materiality; persistence; reversibility; recoverability effect; restoration-pathway effect; dependencies; uncertainty; temporal basis.

Extinction Condition States

VXCM may recognize condition states such as:

1. Potential Extinction Condition

A condition has been identified but its extinction relevance has not yet been established.

2. Non-Material Condition

The condition exists but does not materially threaten recoverability of the governed value.

3. Conditional Extinction Condition

The condition becomes extinction-relevant only under specified circumstances.

4. Material Extinction Condition

The condition materially threatens recoverability and should enter extinction-threshold assessment.

5. Compound Extinction Condition

Multiple interacting conditions collectively create material extinction relevance.

6. Indeterminate Condition

Available information is insufficient to establish the condition's relationship to recoverability.

These are governance states rather than universal technical classifications.

Value Dimension Boundary

A Value Object may support multiple value dimensions.

VXCM must therefore identify which value dimension is affected.

A condition capable of extinguishing:

productive value

may leave:

material value

historical value

informational value

or

strategic value

untouched.

Therefore:

Extinction Condition for One Value Dimension $\neq$ Extinction Condition for Every Value Dimension.

Context Boundary

A value may cease to be recoverable in one context while remaining recoverable in another.

For example:

A technology may permanently lose commercial value in one market but retain operational value in another environment.

VXCM therefore requires contextual extinction conditions to remain bounded to their relevant context.

Contextual Non-Recoverability $\neq$ Universal Non-Recoverability.

Temporal Boundary

The time at which a condition is evaluated matters.

A condition that appears terminal today may become reversible through future capability.

VXCM does not require speculation about every imaginable future technology.

But it does require known or reasonably established restoration pathways and material uncertainty to remain visible.

The Module therefore distinguishes:

currently unavailable recovery

from

governed evidence of terminal recoverability loss.

VXCM Boundary with Value Degradation

Value Degradation governs declining value.

VXCM governs conditions that may indicate movement toward terminal loss of recoverability.

Therefore:

Severe Degradation $\neq$ Extinction Condition Automatically.

VXCM Boundary with Value Nullification

Value Nullification governs the transition into zero value.

VXCM determines whether conditions surrounding zero value materially threaten recoverability.

Therefore:

Nullification $\neq$ Extinction Condition by Definition.

VXCM Boundary with Value Restoration

VRSS governs recovery where restoration potential remains.

VXCM becomes relevant where conditions raise a legitimate question about whether that restoration potential is approaching its terminal boundary.

Therefore:

Restoration Difficulty $\neq$ Extinction.

Loss of Legitimate Restoration Potential $\rightarrow$ Extinction Consideration.

VXCM Boundary with Extinction Threshold

VXCM identifies and qualifies the conditions relevant to possible extinction.

It does not determine where the terminal boundary lies.

That subsequent function evaluates whether the qualified conditions satisfy the governed extinction threshold.

Therefore:

Extinction Condition $\neq$ Extinction Threshold.

Minimum Implementation Framework

A minimum VXCM implementation should establish:

1. Governed Value Identification

Identify the specific value being evaluated.

2. Current Condition Identification

Establish the current value condition and relevant lifecycle position.

3. Potential Extinction Condition Identification

Identify conditions potentially affecting recoverability.

4. Condition Classification

Classify the nature of each condition.

5. Materiality Assessment

Determine whether each condition materially affects the governed value.

6. Persistence and Reversibility Assessment

Determine whether the condition is temporary, persistent, reversible, uncertain, or structurally irreversible.

7. Recoverability Relevance Assessment

Determine how the condition affects recovery of the same value.

8. Dependency Assessment

Identify compound and dependent conditions.

9. Condition Qualification

Determine which conditions legitimately qualify as governed Extinction Conditions.

10. Threshold Transfer

Transfer qualified Extinction Conditions into extinction-threshold assessment.

Governance Outputs

VXCM may produce:

Potential Extinction Condition Record Extinction Condition Classification Condition Materiality Assessment Condition Persistence Assessment Condition Reversibility Assessment Recoverability Relevance Assessment Restoration Pathway Impact Record Condition Dependency Record Compound Extinction Condition Record Condition Uncertainty Record Extinction Condition Qualification Governed Extinction Conditions Record

Enterprise Application

VXCM enables organizations to identify when serious value loss has become significant enough to justify formal extinction governance without prematurely declaring the value extinct.

This is particularly useful where organizations manage:

aging industrial assets; legacy technologies; dormant intellectual property; distressed infrastructure; inaccessible data; obsolete platforms; discontinued capabilities; stranded assets; abandoned projects; impaired digital assets; long-term asset portfolios.

The Module creates a controlled transition between:

“This value is in serious difficulty.”

and

“There is now a legitimate basis to test whether this value is approaching extinction.”

That distinction can prevent both premature write-off and indefinite preservation of unrealistic restoration assumptions.

Use Case 1 — Industrial Production System

A specialized production system reaches zero productive value after years of degradation.

The original manufacturer no longer supports the equipment, several components are unavailable, control software is obsolete, and an attempted restoration program has failed.

VXCM does not conclude:

Value Extinct.

Instead, it identifies and qualifies the relevant conditions:

Component Unavailability → Technological Obsolescence → Control-System Dependency Loss → Failed Restoration Attempt → Alternative Recovery Pathway Assessment → Recoverability Relevance → Governed Extinction Conditions

If alternative reconstruction, modernization, or conversion remains legitimately possible, the productive value may remain recoverable.

If the interacting conditions materially threaten all credible restoration pathways, they become a governed basis for extinction-threshold assessment.

Use Case 2 — Historical Digital Dataset

A historical dataset remains physically stored but essential metadata, decoding information, contextual documentation, and provenance relationships have been progressively lost.

The raw files still exist.

VXCM therefore does not equate:

Data Existence

with

Informational Value Recoverability.

The Module identifies:

Metadata Loss → Context Loss → Decoding Dependency Loss → Provenance Discontinuity → Reconstruction Possibility → Condition Interaction

→ Recoverability Relevance

If alternative records or reconstruction methods remain available, extinction conditions may remain conditional.

If the combined information losses materially threaten the possibility of reconstructing the original informational value, the conditions may qualify for downstream extinction-threshold assessment.

Foundational Principles

Extinction Condition ≠ Value Extinction.

A terminal-looking event does not automatically create terminal value loss.

Extinction conditions must be evaluated against a specific governed value.

Object-level conditions do not automatically apply to every value associated with the object.

One failed restoration pathway does not establish loss of recoverability.

Failed ReMonet™ may inform extinction governance without proving extinction.

ZVA™ does not constitute extinction by definition.

Dormancy does not constitute extinction by definition.

Obsolescence does not constitute extinction by definition.

Economic irrationality to restore does not necessarily mean the value is extinct.

Persistent conditions are not automatically permanent conditions.

Irreversible conditions do not automatically extinguish every value dimension.

Compound conditions may create terminal relevance even where individual conditions do not.

Uncertainty must remain visible before terminal conclusions are made.

VXCM establishes whether the conditions for legitimate extinction consideration exist; it does not determine that extinction has occurred.

Canonical Closing Statement

Value Extinction Conditions Module establishes the first operational governance layer of the Value Extinction Standard by identifying, defining, classifying, contextualizing, and qualifying the conditions that may materially threaten the recoverability of previously existing value. Through governance of condition materiality, persistence, reversibility, recoverability relevance, restoration-pathway effects, dependencies, compound conditions, dimensional boundaries, contextual boundaries, temporal factors, and uncertainty, VXCM prevents degradation, dormancy, zero value, obsolescence, economic infeasibility, failed restoration, or object-level

change from being prematurely interpreted as Value Extinction. The Module establishes the governed conditions necessary for extinction-threshold assessment while preserving the fundamental VFM® distinction that an extinction-relevant condition is evidence of possible terminal value loss, not proof that the value lifecycle has already ended.

Modul2

Governed Links

→ Value Extinction Standard (VXSS)

→ VFM® Architecture Map

→ Next: Value Extinction Threshold Module (VXTM)

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